pyplant workbench | Start stack | Stop | Open HMI | 4/4 services | alarms 0 | faults0

# Break the plant. Watch who notices.

A live two-tank unit with its PID, PLC, historian and operator HMI — and four detectors that compete to find a leak the alarms never see.

TWO-TANK PROCESS
SOFT PLC · 100 ms SCAN
MODBUS · OPC UA · DUCKDB
ISA-18.2 ALARMS

**BREAK SOMETHING**

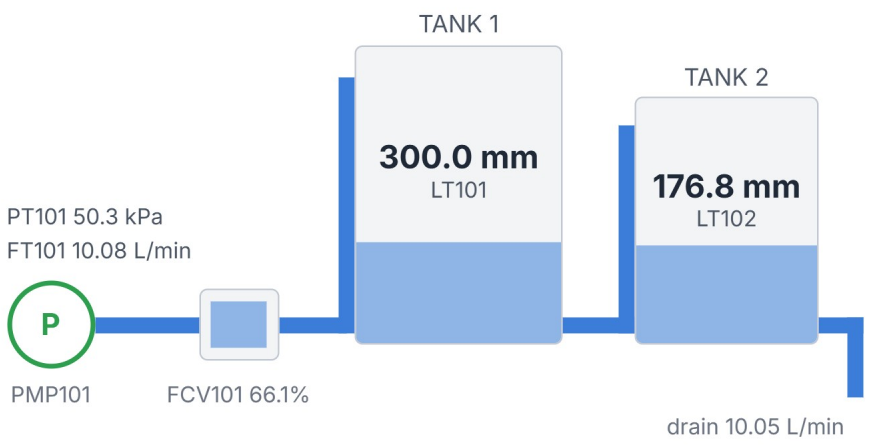
- Leak in tank 1
- Slow leak (15 min ramp)
- Sticking valve
- Worn pump
- Fouled interconnect
- LT101 reads 25 mm high
- LT101 frozen
- Clear all faults

**ACTIVE FAULTS**

Plant is healthy.

Neither the HMI nor any model is told. You find it by watching the plant.

**UNIT 1** — AUTO · healthy




| | | |
|---|---|---|
| FT101 10.08 L/min | FT102 10.05 L/min | PT101 50.3 kPa |
| ZT101 66.1 % | SCAN 3.70 ms | OVERRUNS 0 |

**LIC101**

| | |
|---|---|
| PV 300.0 mm | SP 300.0 mm |
| OP 66.1 % | Error 0.0 mm |

Setpoint 300.0 Write

Mode Auto Manual

**WHAT THE LOOP DOES**

Unit 1 holds 300 mm in automatic. Pick a fault on the left and watch the loop hide it.

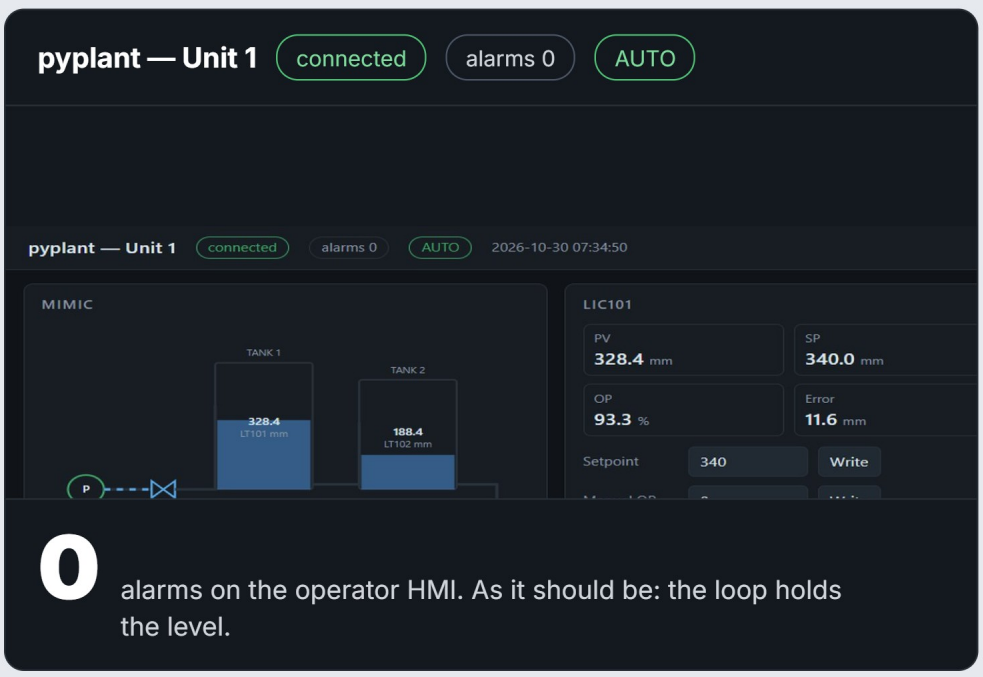
**RESULTS · QUICK DATASET · 14 RUNS, 16 814 SAMPLES**

| model | detected | median delay | false alarms/day |
|---|---|---|---|
| PCA / MSPC (selected) | 87.5% | 43 s | 0.00 |
| Autoencoder | 87.5% | 43 s | 0.00 |
| Z-score limits | 62.5% | 341 s | 0.00 |
| Isolation forest | 37.5% | 354 s | 0.00 |

**BY FAULT MODE · EXPECTED CHANNEL**

–

For leak, pump wear, stiction and fouling the expected column names the physics channel the fault should implicate; a match is an isolated fault.




**0** alarms on the operator HMI. As it should be: the loop holds the level.

**4** live services · plant, PLC, historian, HMI

**7** one-click fault presets

**4** detectors trained and compared

**0** alarms on the leak

```
$ conda activate pyplant
$ python -m pyplant.workbench.app --workspace ./work/manual
→ http://127.0.0.1:8050 · HMI on 5405
```

**Seventeen steps. One leak. Four detectors.**

quickguide.pdf · main.pdf · walkthrough.mp4

Modbus 5400 · PLC 5402 · OPC UA 5403 · historian 5404